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Accounting Information Systems (AIS)
CUAC 204

Question 1

a) **Discuss the difference between internal and external users information, their need and demands on an information system. (10 marks)**

Answer:

INTERNAL USERS

Management

They use the information for planning that is the development of short term and long term plans. Management is concerned with the summarised information I.e reports and draft financial statements.

Supervisors

Controlling activities that involve motivating employees and evaluation of people and other resources within the organisation. Supervisors are interested in detailed information so that they give orders and supervise effectively.

Internal auditors

They use the information that enables to access operational and internal controls . These people are mostly interested in transanction processing systems and information flow charts so that they can establish proper internal controls.

employees

They use the information in order to carryout their daily duties. They need information that is as detailed as possible inorder to reduce errors. They are most information need is turn around documents.

External users

Tax accountants must understand the client's AIS adequately to be confident that it is providing complete and accurate information for tax planning and compliance work.

Potential investors

They want information that will enable them to access the feasibility of investing their funds in a particular entity or to withdraw their funds and invest it else where. Information can be obtained from the SOFP, SOCI and SOCE

Creditors

They want to access the ability of the company to settle its debts. They are interested information that varifies the liquidty position of the business as well as existence and ownership of assets that may want to be used as colleteral.

Debtors

They are mainly concerned about the going concern of a company. They will look for management assertions as to the ability of the firm to continue operating into the forseable future.This will assure them if the company is able to supply them with raw materials in the future(continued business).

b) Distinguish between Accounting Information Systems (AIS) and Management Information Systems (MIS). (10 marks)

AIS is concerned about day to day running of the business with subsystems such as Transaction Processing System (TPS).

MIS is mainly used for strategic planning which involves identifying a company's long run goal and developing plans for achieving these goals.

AIS records financial transactions which are economic events that affect assets and equities of a firm and is measured in monetary terms for instance sale of goods to customers cash movements etc.

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MIS goes beyond AIS in that it records financial transactions and non-financial transactions.

AIS provides information for internal and external use

MIS MIS principally provides information to internal parties.

AIS is used as a tool for short term decision making.

MIS gives data for long term decision making and control.

AIS uses historical data.

MIS is futuristic that is decisions are made based on projections.

AIS supports stewardship management with information contained in the financial statements and responsibility report. Information is about how resources are utilised and safeguarded.

MIS is used as a yardstick for measuring performance. Managers make future projections and hence actual results are compared with the projections made.

AIS provide information that conforms to Generally Accepted Accounting Principles (GAAP).

MIS - Albright (2006) goes on to say it is produced in a variety of formats that meet the particular information needs of a company's managers.

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Question 2

Discuss the importance of accounting independence in Accounting Information Systems. Give where this concept is important. (20 marks)

Definition of terms

Accounting independence: The state where the accounting department is free from any bias or undue influence

Accounting information systems: A combination of components in the organisation that deal with the collection, processing, management and generation of mainly economic transanctions expressed in monetary terms.

1. **Reliability**

Accounting information needs to be independent so that it can be relied upon by both internal and external users. According to James A Hall, reliabilty is the abilty of information to be trusted to influence a certain individual's actions. If the accounting department is not independent, it may be difficult to rely on such information

2. Decision making

Accounting information needs to be independent so that it can be used in decision making. The decision making that is done in accounting affect all department so it is necessary for the information to be independent to facilitate fair decision making

3. Neutrality

Accounting information needs to be independent so that it can maintain its neutrality. Neutrality is a prerequisite for faithful representation of financial information. If information is affected or influenced by other departments, then it loses its neitrality

4.Credibility

Accounting informations needs to be independent so that it can be credible when it is presented. Reports from the accounting department should be validating and enhacing the appeal of companies

5. Standard requirements

A lack of independence may lead to a failure to fulfill professional requirements to obtain enough evidence. Failure to do this undermines the credibility of the accountancy profession and standards it enforces

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6. Proper support

The accounting department supports other departments with information but does not actively participate in them. This calls for independence for this support to serve the true goal for which it is confered.

7. Allocation of resources

The accounting department needs to be independent so that it can allocate resources efficiently. This is the department which is responsible for the overal custody and destribution of resources in the organisation so it needs to be independent for an objective allocation

8. Proper maintainance

The accounting depatment must be separate and independent of the functional areas maintaining resources. This is because even the departments such as the stores department are monitored

by the accounting department so independence is very important.

9.Internal controls

the accounting information depatment needs to be independent so that it can establish proper internal controls. This will avoid situations in which other departments have access to the review of their own work

However

However the independence of the accounting department may bring forward problems such as, costs, tunnel vision and autocracy

1. Costs

Setting up an independent accounting departments will require setting up boundaries and interferces which may be expensive to put in place when considering the cost benefit analysis

2.Tunnel vision

In practice, the accounting department can not be pratically independent since this may lead to the department only thinking of factors that affect it which is called tunnel vision. Practically the accounting department should liason with the I.T department for the establishment of good information systems.

3. Autocracy

If the accounting department is compeletely independent, It may end up making unilateral decisions without consulting the views of other departments that may be affected by the same information.

Conclusion

In conclusion, the independence of the accounting department is very important despite the fact that the independence of the department may end up bringing up problems.

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Question 8

Why should an auditor be involved in the system design process (15 marks).

1. Establishment of controls

The auditor needs to be invloved in the system design process so that he can oversee and suggest internal controls in the system that are effective. In the process, the auditor will be responsible for suggesting the boundaries and interfaces that are needed in the system.

2. Audit trail

The auditor is needed in the system design so that he can understand the trails and paths that are followed by documents as they move through the organisation. This is very important in the organisation since the auditor will use the information in the tracking and tracing of transactions

4. System check

Before the enron scandal, Accountants left the job of systems design to the IT guys. However because the enron scandal was partly because of information system failures and loop holes, the concept of auditors being system checking was introduced so that the auditors can satisfy themselves with the functionalities of the system.

5. IT auditing

Since auditors now use Computer assisted auditing, They need to be involved in the system design process so that they can identify the ideal auditing system that is suitable for a particular firm to reduce manual processng and auditing

6. Assuarance

The auditor's Job is to give assuarance. The auditor needs to be fully aware and knowledgebale of the information systems that are used by a certain firm so that they can give a proper assuarance that can be relied upon.

7. Navigation

If the auditor is included in the systems development, he can easily navigate through the the departments of the organisation and know where the important files of the organisation are kept.

Arguments against:

Self review threat

If the auditor is included in the design of an information system, his independence may end up being compromised since along the way he may have to evaluate and audit the same system that he helped build. The provision of consulting and assuarance services is prohibited by the Surbanes Oxley Act.

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Familiarity threat

For the auditor to be included in the systems development, he needs to interact and socialise with IT designers and information designers of an organisation since it may take time to build such a system. After a long time, his independence may be compromised as relationships may build along.

Question 20

a) "A system is a set of parts coordinated to accomplish a set of goals" West Churchman. Give three main characteristics of a system. [6 marks]

Component parts/ tangible features .

These imply structure and order. The components are arranged to help achieve objectives for example a computer system is designed around an input device, a central processing unit, an output device and one or more storage units. When linked together they work as a whole system to produce information.

Process.

This is whereby the component parts co-ordinate in a defined way. Process refers to the manner in which each components function with other components of the system. For example, in a computer system the CPU must interact with the input device to solve a problem, in turn the main memory holds the program and data that the arithmetic unit uses for computation. This means that the parts of the system depend on each other and they are coordinate and linked together according to plan.

Goal

These are the objectives which the components parts are coordinated. The objectives may be real or stated. In most organizations, they state an objective and operate to achieve it

Question 6

"Unlike systems that occur naturally, Accounting Information System, must be created."

Give 5 qualities of a successful system and how each quality may be achieved (20 marks).

Definition of a system

A system is a set of parts coordinated to accomplish certain objectives or a group of interrelated multiple components that save a common purpose

Accounting information system is an information system with set of procedures by which data are collected, processed into information, and distributed to users.

A system is said to be successful when it achieves its objectives and it must have the following qualities

Quality	Achieved by
1. Correct and reliable information	implement adequate internal controls 1)through segregation of duties 2)use of confirmations and inquiries

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	3)independent verification that is to check to assess if the information produced is correct -maintenance, servings and reviewing of the system at intervals -Correct procedures of recording transactions
2. Timely information	Chose a processing method suitable to user's needs. For example use of online processing method as it processes transactions at the moment the event occurs
3. User satisfaction	-user input in setting policies -user support for project -activate user participation in project -user responsibility for the system -training programs to familiarize the user with the new system
4. Satisfy the organization's needs a)current b)future	a) Getting steering committee approval for new system development b) Have a long range system master plan this governs the objectives of the organization.
Adopt to technological changes 5. Reasonable time for development	-higher creative and innovative employees -have systems analyst and expertise responsible for research and development. For example pastel use can easily change from version 8 to 17. -properly define the scope of the system -use project management techniques

6.

It is logical to conclude that for a system to be successful it must achieve its objectives and have an overall vision. However, there is no system which is completely successful.

b) What are subsystems and super systems in accounting information system and why should an accountant be knowledgeable about these systems? [8marks]

Sub-system.

A subsystem is a system that exists in another system; its existence depends upon the existence of a super system. A sub system contributes to the proper working of the entire system. For example, an auto mobile system consists of many sub-systems like acceleration system, fuel injection system and the braking system. Every sub-system is responsible for the proper functioning of the entire auto mobile system.

Super system

A super system is a system responsible for monitoring the overall working of its sub-systems therefore it contains many sub-systems. It decides the constraints and resources to be put on its sub-systems. For example; the central government system which has the state government under its control which form the

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sub-system.

Usefulness

Enables the accountant to navigate suitable system software;

the knowledge will enable the accountant to confirm and record actual business transaction, describe economic events, perform activities timely and sufficiently, sort the economic property in the financial reports.

Enables accountants connectivity

to access the barriers for implementation of financial control of owners' connectivity with financial managers. The accountant will seek ways to improve and reach maximum operations if he reviews that the sub-system and super systems and becoming obsolete.

Enables Evaluation

The knowledge enables accountants to evaluate the business performance correctly and predict the prospects of the enterprise.

Analysis of systems theory and corporate financial control;

knowing the characteristics of systems and there organic in nature, an accountant can promote the smooth development of corporate financial activities.

c) Why is it necessary to create boundaries and interfaces when designing accounting information systems? [6 marks]

To enable efficiency allocation of resources to each sub-system.

To make it easier for accountants and auditors to review on the effectiveness of internal controls.

To enable the achievement of almost every characteristic of a successful system since the components parts, processes and goals will be established against each boundary.

Boundaries simplify the process of organizing subsystems so as to reduce the number of interconnections, which is a potential interface for communication among sub-systems.

Interfaces allow accountants to control the flow of information in a system which is a prerequisite for batch processing which is favoured by accountants

It allows accounants to filter information that is not neccessay during data collection. This will reduce labour required on data management and processing

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Question 10

Discuss what is meant by the following:-

Give examples to expound your explanations (20 marks)

a) Closed systems

A closed system is a system that does not interact with the environment. There are no interfaces to connect the system to other systems or the environment at large. This means that the effects on the outside of its boundaries and the environment has no effects on the process within the system. A research and development (R&D) department within a computer organisation may be set up as a closed system with no interaction outside the division. Such a system is a theoretical concept rather than being a practical concept.

ADVANTAGES

There are no disturbances in the system.

It is highly effective and efficient.

LIMITATIONS

Can hamper growth since the flow of information stays within the system.

Has no chance to interact with or build on knowledge from the outer world.

It does not adapt to technological changes.

It practically does not exist because it lacks the need for interaction.

b) Relatively closed systems

A releviley closed system is a system that reacts with its environment in a known and controlled way. It is semi permeable.The interactions consist of system inputs (if they flow from the environment to the system) and outputs (which are interactions flowing in the opposite direction). The system contains interfaces with the environment and controls the effects of the environment on its process. It interacts with its environment in a restricted way that does not eliminate but limits its susceptibility to the environment.For example, labour negotiations can limit the effects of strike.

ADVANTAGES

Able to meet user needs as they are able to take inputs for development purposes.

It is flexible – adapts to technological changes.

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It allows for a reasonable level of sharing and exchange of information.

Minimises the impact of disturbances as there are control measures.

DISADVANTAGES

It is not reliable – for it is prone to disturbances from the environment

If the control points fail to operate, it may have information overload or information losses

The interfaces may be dificult and expensive to establish.

c) Open systems

Is one in which the systems interaction with the environment is not controlled.Has many interfaces with its environment ,that is ,system that interacts freely with its environment, taking input and returning output. Contains subsystems which help in converting input into output.

It permits interactions across its boundary , it receives inputs from and delivers outputs to the outside.System designers anticipate the things that can go wrong in the environment and create processes and interfaces to control them. For example, media and internet.

Advantages

Have flexible boundaries since it interacts with the environment.

Promote effective problem solving.

Subsystems help in resolving conflicts as well as allocating resources.

Disadvantages

The system has disturbances of uncontrolled inputs that affect the processed resources within the system.

The system can suffer from information overload

It is difficult to trace sources of error in the system

d) Feedback control systems

It is a system where a portion of the system is retained as an input to the system.For example, in accounting system ,revenue output by other goods can be used as capital in costing and variances

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can be output that can be used in decision making. It is designed to provide feedback

Uses interfaces to permeate information bottom up and top down.

Uses the function of a prescribed relationship between the output and reference input to control the process of resources in that system.

The difference between the output of the process under control and the reference input is amplified and used to control the process so that the difference is continuously reduced

Advantages

Can be used for decision making.

Provide the feedback which help the system to obtain its goals and enhance employee motivation.

Provide management with meaningful information on how effective its planning effort was.

The effect of unwanted disturbances can be effectively reduced

Disadvantages

The time the manager may get the information , the damage might already have been done thus making feedback irrelevant

It is expensive to set up and the cost of feedback is first manifested in the increased number of components and the complexity of the system.

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Question 11

Diagrammatically present the four transaction cycles identified by most auditors and briefly explain what is involved in each cycle. (20 marks)

TRANSACTION CYCLES

Four transaction cycles process most of the firm's economic activity: the cash purchases and disbursement expenditure cycle, the payroll and fixed assets cycle, the conver- sion cycle, and the revenue cycle. These cycles exist in all types of businesses—both profit-seeking and not-for-profit types.

For instance, every business

(1) incurs expenditures in exchange for resources (cash purchases expenditure cycle),

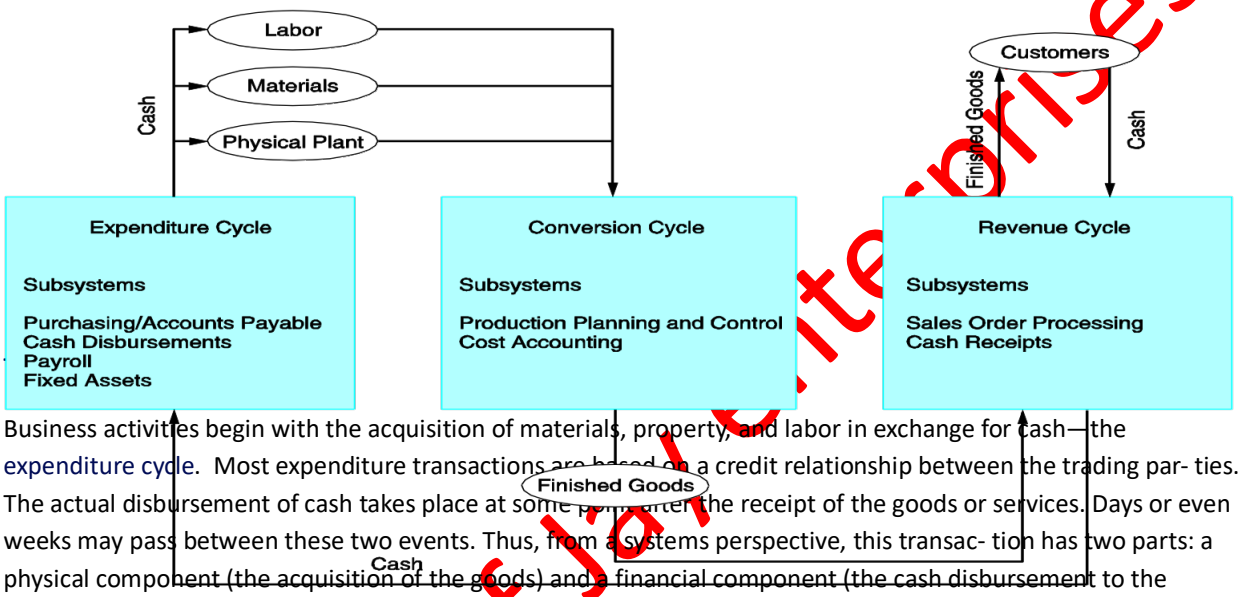
(2) pays out money for services rendered by employees and acquisition of fixed assets (the payroll and fixed assets cycle),

(3) provides value added through its products or services (conversion cycle),

(4) receives revenue from outside sources (revenue cycle).

In each of these cycles, there are two components namely the financial and physical component.

Relationship between Transaction Cycles



Business activities begin with the acquisition of materials, property, and labor in exchange for cash—the expenditure cycle. Most expenditure transactions are based on a credit relationship between the trading par- ties. The actual disbursement of cash takes place at some point after the receipt of the goods or services. Days or even weeks may pass between these two events. Thus, from a systems perspective, this transac- tion has two parts: a physical component (the acquisition of the goods) and a financial component (the cash disbursement to the supplier). A separate subsystem of the cycle processes each component. The major subsystems of the cash purchases and cash disbursements expenditure cycle and the payroll and fixed assets expenditure cycle.

Purchases/accounts payable system. This system recognizes the need to acquire physical inventory (such as raw materials) and places an order with the vendor. When the goods are received, the pur- chases system records the event by increasing inventory and establishing an account payable to be paid at a later date.

Cash disbursements system. When the obligation created in the purchases system is due, the cash disbursements system authorizes the payment, disburses the funds to the vendor, and records the transaction by reducing the cash and accounts payable accounts.

Payroll system. The payroll system collects labor usage data for each employee, computes the payroll, and disburses paychecks to the employees. Conceptually, payroll is a special-case purchases and cash disbursements system. Because of accounting complexities associated with payroll, most firms have a separate system for payroll processing.

Fixed asset system. A firm's fixed asset system processes transactions pertaining to the acquisition, maintenance, and disposal of its fixed assets. These are relatively permanent items that collectively of- ten represent the organization's largest financial investment. Examples of fixed assets include land, buildings, furniture, machinery, and motor vehicles.

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The Conversion Cycle

The conversion cycle is composed of two major subsystems: the production system and the cost account- ing system. The production system involves the planning, scheduling, and control of the physical product through the manufacturing process. This includes determining raw material requirements, authorizing the work to be performed and the release of raw materials into production, and directing the movement of the work-in-process through its various stages of manufacturing. The cost accounting system monitors the flow of cost information related to production. Information this system produces is used for inventory valuation, budgeting, cost control, performance reporting, and management decisions, such as make- or-buy decisions.

Manufacturing firms convert raw materials into finished products through formal conversion cycle operations. The conversion cycle is not usually formal and observable in service and retailing establish- ments. Nevertheless, these firms still engage in conversion cycle activities that culminate in the develop- ment of a salable product or service. These activities include the readying of products and services for market and the allocation of resources such as depreciation, building amortization, and prepaid expenses to the proper accounting period. However, unlike manufacturing firms, merchandising companies do not process these activities through formal conversion cycle subsystems.

The Revenue Cycle

Firms sell their finished goods to customers through the revenue cycle, which involves processing cash sales, credit sales, and the receipt of cash following a credit sale. Revenue cycle transactions also have a

physical and a financial component, which are processed separately. The primary subsystems of the reve-

nue cycle, which are the topics of Chapter 4, are briefly outlined below.

Sales order processing. The majority of business sales are made on credit and involve tasks such as preparing sales orders, granting credit, shipping products (or rendering of a service) to the customer, billing customers, and recording the transaction in the accounts (accounts receivable, inventory, expenses, and sales).

Cash receipts. For credit sales, some period of time (days or weeks) passes between the point of sale and the receipt of cash. Cash receipts processing includes collecting cash, depositing cash in the bank, and recording these events in the accounts (accounts receivable and cash).

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Question 12

Compare and contrast batch processing and online processing, outlining the meaning, advantages and disadvantages of the two. Which one would you recommend for a manufacturing entity where the accounting function is decentralized? (20 marks)

http://slideplayer.com/slide/6014166/

Batch processing

- Source documents are grouped into batches andcontrol totals calculated.Periodically (say after some minutes, days, weeks or months), the batches are entered into a computer system, edited, sorted and stored in a temporary file.
- The temporary file is run against the master file to update the master file.Output is displayed/printed along with error and transaction reports.
- A batch is a group of similar transactions that are accumulated to be processed together.Under batch processing, the data is collected before any processing can take place; it is then entered into the system in one go (a batch of data) and processed.Batch processing therefore is when the computer does not carry out any processing/ does not produce any output until all the inputs have been collected in.

Examples of Batch processing

- Customer billing
- Payroll systems
- Preparation of financial statements

Payroll systems

The hours worked by each employee need to be collected first.These will be fed into the computer hebdomadal, weekly or monthly for the computer system to calculate pay and produce pay slips.

Input hours worked Master file Transaction File Calculate pay
Fig 1. system flow chart to show batch processing in payroll systems. (Draw own diagram)

Calculate pay

Pay slips

Advantages of batch processing

- Organizations can increase efficiency by grouping large numbers of transactions together rather than processing each separately (economies of scale).
- Batch processing allows the system to transfer resources for other purposes when the system is not busy.It is possible to perform repetitive tasks on a large number of pieces of data rapidly without needing the user to monitor it.

Online Processing Data is processed by the system as it is entered.

- The master file is then immediately updated.
- A subset of online processing is real time processing.

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- Under real time processing, response time and processing speed are enhanced to effect output before the next input is done.

Examples of online processing

- Airline booking systems
- Patient monitoring systems
- GamesOrder processing

Order Processing An order is entered into the system.

- The system checks whether there are any inventories at hand in order to accept the order. If there are no inventories the system stores the order as an open order in the open order file, to be closed when the order is met, and then prints/displays open order report. If there are inventories, the system accepts the order and prints/displays the closed order report.The master file is updated each time an order is closed, to show the implied reduction in inventory.

Input order Inventory Master Authorize File order Open orders Closed
Fig 2. system flow chart to show online processing in airline order systems.(Draw own diagram)

Open orders

ClosedordersOpenorders

Advantages of online processing

- It enables the user to input data and get the results to the processing of that data immediately.
- Master file is updated immediately which enables the provision of timely and relevant information.

Differences between online and batch processing
Online processing

A lag exists between time when the economic event occurs and when it is recorded.Generally fewer hardware, programming and training resources are required.Certain records are processed after the event to avoid operational delays.

Processing occurs when the economic event takes place.More resources (hardware) and dedicated processing capacity and quality are required.All records pertaining to the event are processed immediately.

Recommendations for processing method in a manufacturing entity where the accounting function is decentralized.

Online processing

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Since the accounting function is decentralized, the entity will find it desirable to use mainly online processing since the master files they share need to be updated as events occur in order to make timely decisions.For example, the accounts receivable master file will need to be updated as soon as credit is granted to a customer in order to avoid granting further credit to undeserving customers.

Batch processing

However, the firm may also employ batch processing in other operations in order to enhance operational efficiency.For example, updating the sales journal file may be done using batch processing during lunch time or any other time when the system is not busy.

Question 9

"It is difficult and tiresome to come up with an accounting information system". Discuss how the use of cycles help in making sense of accounting applications, give examples and flow charts of such cycles in your discussion. (20 marks)

in book

Question 26

(a) Explain how the concepts of responsibility, accountability and liability apply in relation to the ethical use of information systems. (6 Marks)

Responsibility means that individuals or organisations accept the potential costs, duties and obligations for the decisions they make in relation to the use and management of information.

Accountability means that mechanisms are in place to determine who took responsible action, and who is responsible. A failure to put these in place means that the actions taken may not be ethical.

Liability extends the concept of responsibility into the area of law. A body of laws exists that permits individuals (or organisations) to recover damages done to them by other organisations, systems or other actors.

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(b) Discuss the main ethical issues that need to be taken into consideration when using a customer relationship management system to collect, store and use customer data. (8 Marks)

1. Privacy
2. Security (Accuracy and confidentiality)

3. Misuse of data
4. equity in access
5. ownership of property

Alt answer: Ethical issues around the collection of customer data for CRM are related to secure collection methods and to the verification of the information. Ethical companies ensure that sensitive information is obtained and processed fairly (as stipulated by the Data Protection Act 1988 & 2003), collected in a secure environment and transmitted back to the databases securely. During data collection, it is critical to verify the identity of the customer and the accuracy of the information being submitted. High security for these functions is costly but ethically necessary.

Once customer data is stored in a company database, the company should adhere to the following principles regarding storage. These are outlined in the Data Protection Act:

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The ethics behind these principles are that the data belongs to the customer and the customer must be able to control his/her data. Customer data should therefore only be stored with the agreement of the customer. Customers should be able to view their data and change it (or ask for it to be changed), and can ask for it to be erased. Companies may not divulge the personal data of customers to a third party, except in ways that are "compatible" with the specified purpose. A key test of compatibility is whether they use and disclose the data in a way in which the customers that supplied it would expect it to be used and disclosed.

Given that much of the customer data for CRM is sensitive, companies must ensure the data is kept private to the maximum extent possible. To achieve this, a company should store the data in a form or in a location that is not generally accessible. The data must only be consulted when necessary for the fulfillment of a CRM task, and only those employees who handle the data to complete the task should be able to access the data.

(c) Outline the type of controls that a company should have in place to avoid misuse of its information assets? (remember control activities)

1. Preventative controls
2. Detective controls
3. Corrective controls

(6 Marks)

Question 32

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(a) List and describe the stages in decision making. (5 Marks)

1. Identification of Goals

One of the most effective decision making strategies is to keep an eye on your goal. This simply means identifying the purpose of your decision by asking yourself what exactly is the problem that needs to be solved? And why does this problem need to be solved? Figuring out what's most important to you will help you make good decisions. When you know the reason why you have making a particular decision; it will better serve you in staying with it, and defending it.

2. Gather Information for Weighing Your Options

When making good decisions it is best to gather necessary information that is directly related to the problem. Doing this will help you to better understand what needs to be done in solving the problem, and will also help to generate ideas for a possible solution.

When gathering information it is best to make a list of every possible alternative; even ones that may initially sound silly or seem unrealistic. Always seek the opinions of people that you trust or speak to experts and professionals, because it will help you to come up with a variety of solutions when weighing all your options for a final decision. You will want to gather as many resources as possible in order to make the best decision.

3. Consideration of Consequences

This step can be just as important as step one because it will help you determine how your final decision will impact yourself, and/or others involved. In this step, you will be asking yourself what is likely to be the results of your decision. How will it affect you now? And how will it affect your future?

This is an essential step because it allows you to review the pros and cons of the different options that you listed in the previous step. It is also important because you want to feel comfortable with all your options and the possible outcome of whichever one you choose.

4. Make and implementation of Decision

Now that you have identified your goal, gathered all necessary information, and weighed the consequences, it is time to make a choice and actually execute your final decision. Understanding that this step can cause some people a lot of anxiety is important because this is where you have to trust your instincts. Although you may still be slightly indecisive about your final decision, you have to take into account how this makes you feel. Ask yourself, does it feel right? And does this decision work best for you now, and in the future? When you answer those questions back, you should feel good about the result.

5. Evaluation of Decision

Once you have made your final decision and put it into action, it is necessary to evaluate the decision and the steps you have taken to ensure that it works. This final step is probably just as important as step

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one, if not more important, because it will help you to further develop your decision making skills for future problems. This step is also fundamental because it may require you to seek out new information and make some changes along the way.

Remember, this step requires some patience and it can also encourage perseverance. Why? Because it may take some time to see the final outcome. Recognizing that if the first decision is not working, you may have to go back to step two and choose another option. Always looking for and anticipating unexpected problems will help alleviate undue stress, if and when a problem occurs.

(b) Distinguish between unstructured, semi-structured and structured decisions. (8 Marks)

Structured decisions	Semi structured decisions	Unstructured decisions
A decision where data, procedure and objectives are known with certainity	A decision where one or two of the elements of data, procedure and objectives are unknown with certainity	A decision where neither the elements of data, procedure and objectives are unknown with certainity
Usually made by operational level managers	Usually made by middle level managers	Usually made by top level managers
There exists a laid out procedure for arriving at a solution	A combination of structured phases, standardized procedures and individual judgement is used to arrive at a solution	Intuation plays an impotant role for arriving at a solution

(c) Define and describe the main capabilities of an Executive Support System. (7 Marks)

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Question 7

The goal of any part of the information system is to help the decision-making process. Robert Anthony developed an approach for categorizing the decision-making activities within an organization. Having identified three levels of such activities, he stated that decision making differs so significantly between these levels that different information systems are needed at each level.

Give these levels and the decision problems for each level, a tabular presentation is encouraged and a write up of the information needs an accounting information system should provide for such decisions to be made. (20 marks)

Triangle diagram Ask sir

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Question 5

The evolution of most accounting systems follows a distinct cyclical pattern commonly called the system development life cycle (SDLC). Discuss in detail the four main components of the SDLC. (25 marks)

Question 3

With an aid of a diagram explain System Development Life Cycle (SDLC) and thee role of the accountant in each stage. (20 marks)

Answer to both questions.

Introduction.

The systems development lifecycle is a multistage process that guides an organisation's management through the development and/or purchase of information systems. It has 5 stages namely Systems strategy, project initiation, in house development, commercial packages and Maintenance and support.

Diagram

General accountant's involvement in the SDLC

Accountants are involved in systems development in three ways.

- First, accountants are users. All systems that process financial transactions impact the accounting function in some way. Like all users, accountants must provide a clear picture of their problems and needs to the systems professional. For example, accountants must specify accounting techniques to be used; internal control requirements, such as audit trails; and special algorithms, such as depreciation models.
- Second, accountants participate in systems development as members of the development team. Their involvement often extends beyond the development of strictly accounting information systems applications. Systems that do not process financial transactions may still draw on accounting data. The accountant may be consulted to provide advice or to determine if the proposed system constitutes an internal control risk.

- Third, accountants are involved in systems development as auditors. Accounting information systems must be auditable. Some computer audit techniques require special features that must be designed into the system. The auditor/accountant has a stake in such systems and must be involved early in their design.

The following is a detailed explanation of the individual stages and how accountants are really involved in the step by step process.

1. Systems strategy

The first step in the SDLC is to develop a systems strategy, which requires understanding the strategic business needs of the organization. This may be derived from the organization's mission statement, an analysis of competitive pressures on the firm, and the nature of current and anticipated market conditions. These needs reflect the organization's current position relative to where it needs to be in the long term to maintain strategic advantage. In addition, project management must consider the information systems' implications pertaining to legacy systems and concerns registered through user feedback. A strategic plan for meeting these various and complex needs, along with a timetable for implementation of selected systems, is produced.

Role of the accountant.

The role of the accountant in this stage is to address critical enterprise-wide issues such as doing investment appraisal on how a new system would provide a better return on existing investment and reduced risk for future investment. The accountant is there to look at the overall financial effect that a new system may have on the organisations goal, vision as well as the overall business strategy.

2. Project initiation

Project initiation is the process by which systems proposals are assessed for consistency with the strategic systems plan and evaluated in terms of their feasibility and cost-benefit characteristics. Alternative conceptual designs are considered, and those selected enter the construct phase of the SDLC. Depending upon the nature of the project and the needs of the organization, the pro- posal will require in-house development, a commercial package, or both.

Role of the accountant.

The role of the accountant here is to be involved in the detailed cost benefit analysis and project proposal appraisal. The accountant must provide an opinion on the feasibility of the project economically I.e the accountant must make decisions that pertain to the availability of funds to complete the project.

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3. In-House Development or Purchase Commercial Software

At this stage, two general options are open to the organization in the construct phase: develop the system in-house or purchase commercial software. some organizations have unique information needs that can be adequately met only through internal development. The in-house development step includes analyzing user needs, designing processes and databases, creating user views, programming the applications, and testing and implementing the completed system. When the nature of the project and the needs of the user permit, most organizations will seek a

precoded commercial software package rather than develop a new system from scratch. The organizations that can implement commercial software accrue a number of advantages. These include lower initial cost, shorter implementation time, better controls, and rigorous vendor testing. All of these benefits translate into cost savings for the user. This process, however, is not without risk. Formal procedures need to be followed to ensure that the user gets a package that adequately meets his or her needs and is compatible with existing systems.

Role of the accountant.

At this juncture, the accountant should provide a consultation as to which option it will follow. Systems that need to meet unique and proprietary business needs are more likely to undergo in-house development. Systems that are expected to support best industry practices may be better suited to the purchased-software option.

5. Maintainance and support. Maintenance involves both acquiring and implementing the latest software versions of commercial packages and making in-house modifications to existing systems to accommodate changing user needs. Maintenance may be relatively trivial, such as modifying an applica- tion to produce a new report, or more extensive, such as programming new functionality into a system. The feedback loops from maintenance to the project initiation and systems strategy steps, respectively, despite these relationships.

Accountant's role

At this stage, the accountant is both the manager of a system and a user. The accountant must solicit feedback from both the external and internal users of the system. The accountant must collect this data as it will be used in future systems development and improvements.

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Question 13

Giving examples, explain how the following systems aid management in decision-making

(i) Expert systems

(ii) Simulation

(iii) Decision Support system

(iv) Information centres (20 marks)

Expert systems

-An expert system is an application of artificial intelligence that emulates knowledge of human expert, solve problems, make decisions and make expertise available to novices.

- The scope of expert system is limited to expert's experience .

- Expert system provides decision support to managers in the form of advise from an expert consultant in a specific problem area.

- Expert system also help management in solving problems which the organization is currently facing in form of, for example, diagnosis and prediction of mechanical failures which is vital for operational managers.

- example: AGEX, a system that gives advice to farmers.

Simulation systems

- Simulation is a broad term that refers to an approach for imitating the behavior of an actual or anticipated human or physical system.

- This involves solving by executing (doing).

- There are three areas covered by simulation which are:

Model design

Model execution

Model analysis

- Simulation helps management in decision making by provision of exact problems faced as a result of the use of the model they have designed.

  – Example: Queing model systems

Decision Support Systems (DSS)

- These are systems that combine data and analytical models or data analysis tools to support structured, semi-structured and unstructured decision making.

- DSS is a computer based information system that help management select solution, reduces costs, Increase profitability and enhances product quality

-DSS comprises of three components that are data management module, model management module and dialog module.

- DSSs are used by management to assist in making decisions on issues which are subject to high levels of uncertainty about a particular problem .They help management in gathering various responses which they would undertake and the likely impact of those actions.

- DSS are intended to provide a wide range of alternatives of information gathering and analytical tools with a major emphasis upon flexibility and user friendliness.

Example: credit rating softwares that are used by many banks to automate decision aids so as to determine the credit worthiness of their clients.

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Question 15

a) **Discuss any four approaches to system changeover. Briefly outlining the advantages and disadvantages, if any, of each approach. [12 marks]**

Answer:

Cold Turkey Cutover

Under the cold turkey cutover approach (also called the big bang approach), the firm switches to the new system and simultaneously terminates the old system. When implementing simple systems.

Advantages

-It is least costly approach.

-It is the easiest approach

Disadvantages

- With more complex systems, it is the riskiest since System errors that were not detected during the walk-through and testing steps may materialize unexpectedly.
- Without a backup system, an organization can find itself in serious trouble.

Phased Cutover

-The phased cutover begins operating the new system in modules. For example a firm might implement a system, starting with the sales subsystem, followed by the inventory control subsystem, and finally the purchases subsystem.

Advantages

- By phasing in the new system in modules, we reduce the risk of a devastating system failure.
- It works perfectly where there is a low degree of intergration, dependancy and continous information sharing in departments

Disadvantages

- the phased approach can create incompatibilities between new subsystems and yet-to-be-replaced old subsystems.
- It does not work where there is high degree of integration, dependency and information sharing of departments in an organisation.

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Parallel Operation Cutover

Parallel operation cutover involves running the old system and the new system simultaneously for a period of time.

Advantages

- The advantage of parallel cutover is the reduction in risk. By running two systems, the user can reconcile outputs to identify and debug errors before running the new system solo
- it allows the user to reconcile the two outputs at the end of the cycle as a final test of the system's functionality.

Disadvantages

- it takes a very period of time for a full system changeover to take place.
- It requires twice the source documents, twice the processing time, twice the databases, and twice the output production which is very costly

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b) Why is a post – implementation review necessary for a new system? [8 marks]

The final step in the implementation phase actually takes place some months later in a post-implementa- tion review.

Reasons for post implementation review

- The objective is to measure the success of the system and of the process after the dust has settled. Although systems professionals strive to produce systems that are on budget, on time, and meet user needs, this does not always happen.
- It provides information on Accuracy of Time, Cost, and Benefit Estimates that were made before the system was implemented
- The post- implementation review of the newly installed system can provide insight into ways to improve the process for future systems.
- It can serve as a perfomance appraisal tool on the workers that were involved in the implementation of the system.
- It helps to check if the users have fully been acquainted with the system.

Question 27

(a) Identify and briefly explain six dimensions of information that affect the quality of decision making in an organisation. (6 marks)

(c) Outline the main business benefits of collaboration, and explain how social networking can be used to help achieve some or all of these benefits. Illustrate your answer with practical examples. (8 marks)

(b) Define the term knowledge management and, using an organisation of your choice, explain what types of knowledge it would have and how such an organisation could benefit from knowledge management. (8 Marks)

(c) Why are knowledge workers so important to the digital firm? (6 Marks)

[Total: 20 Marks]

Question 25

Describe briefly, what is meant by ANY FIVE of the following:

(a) Executive Support Systems

Executive Support System (ESS) is a reporting tool (software) that allows you to turn your organization's data into useful summarized reports. These reports are generally used by executive level managers for quick access to reports coming from all company levels and departments such as billing, cost accounting , staffing, scheduling, and more.

In addition to providing quick access to organized data from departments, some Executive Support System tools also provide analysis tools that predicts a series of performance outcomes over time using the input data. This type of ESS is useful to executives as it provides possible outcomes and quick reference to statistics and numbers needed for decision-making.

(b) Denial of Service attacks

(c) Customer Relationship Management Systems

(d) Open Source Software

(e) Privacy rights under the Data Protection Act

(f) M-commerce

(g) The Internet and Competitive Advantage

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Question 4

a) Contrast the advantages and disadvantages of CASE tools. (15 marks)

b) What should determine whether CASE is used? (5 marks)

Question 16

Give and explain the four main system analysis steps in the study of an accounting information system. [20 marks]

Question 22

a) As an accountant of an organization, discuss why it might be necessary to initiate systems analysis. [6 marks]

b) Giving reasons for your answer, discuss the accounting system you would recommend to a decentralized organization between a computerized accounting information system and a manual accounting information system. [14 marks]

Question 17

Discuss any five elements that one would need in transaction processing. [20 marks]

Question 19

a) Briefly discuss the envisaged benefits by the Government of Zimbabwe for introducing fiscalised cash register for taxpayers who are registered for Value Added Tax (VAT) purposes in light of changes in technology and accounting information systems in general [10 marks] .

Question 23

RECYC Ltd is a medium sized recycling consultancy firm, which is currently experiencing rapid growth. This is evidenced by increased turnover. It has been able to develop a range of new consultancy and specialist business advisory services that it offers to its growing customer base. To cope with these developments, several organisation-wide initiatives have been launched over the past two years.

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The existing financial systems are struggling to cope with these developments, but replacement software is due to be installed within the next six months. The new system was justified partly because it could reduce costs, although precise details have not been given. The application software selected does not fit existing business processes exactly. However, it has the clear advantage of giving RECYC access to an industry best practice system and is identical to that used by all its main competitors and some of its clients.

A three-person Project Steering Group has recommended that a phased approach to introduction should be used and has undertaken most of the project planning. A programme of events for implementing the system has been agreed, but is not yet fully operational. This group has not met for a while because the designated Project

Manager has been absent from work through illness. You are Head of RECYC's Central Support Unit. You also serve on the Project Steering Group.

A Directors' meeting is due to take place soon. The firm's Managing Director has asked you to prepare and present a PowerPoint presentation to the other Directors on project management and implementation issues. You understand that the Directors are conscious that system implementation represents a form of further organisational change. They are asking questions about the approach that will be taken to the introduction of the new system, likely changes to practices, critical areas for success, system testing, support after implementation, system effectiveness, etc.

REQUIRED:

(a) List and describe the project management activities and variables addressed by your steering group. (8 marks)

(b) Describe the available options for the systems implementation and explain why a 'phased' approach to introducing the system is, in this case, more suitable. (7 marks)

(c) Name and describe the methods you are using to evaluate this information systems project and outline the limitations of financial models for establishing the value of information systems. (10 marks)

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Question 24

A teacher training college in the UK has recently expanded very rapidly, introducing a number of programmes and increasing student intake capacity. It has also recently computerised its teaching and assessment processes based on a customised version of some off-the-shelf software. The Library had already been computerised a couple of years ago using a highly functional library automation system that includes multiple catalogs, authority control and so on. Accounts and payroll are handled by computers in the Finance and Accounting department, using popular accounting and payroll software packages.

In order to achieve further improvements in efficiency, the college's management board asked a team of consultants to assess the suitability of its existing technology infrastructure and information systems to meet its strategic objectives. The consultants advised that the college is burdening itself with too many independent information systems. They strongly favoured an integrated information system on the lines of an enterprise resource planning (ERP) solution for a business organisation.They were concerned that adequate consideration may not have been given to all the areas of operation of the college, or to the strategic oversight of these. In particular they were concerned about the lack of automated procedures for the procurement of everything from laboratory equipment to travel tickets; the lack of integration of the Student Records department with Finance & Accounting (especially in relation to fee payments); and the marketing of the college to potential new students.

The consultants' report also noted that the directors of the Finance and Accounting department and the Library were both reluctant to change from their existing information systems. The directors highlighted the fact that their departments had recently spent time and money investing in systems that were perfectly adequate for their needs and suited their operating practices.

While the college's Information Technology department are technically capable of implementing an organisation wide ERP system, they are under-staffed and are highly dependent on key individuals. Management in the department are also concerned that they do not at this time have an adequate hardware infrastructure to support uninterrupted, secure operation of the options being proposed. In fact there has been a history of service outages over the previous six months, all of which were caused by technology failures and some of which resulted in data loss.

Given that management identified 24 hour online access to library resources, e-learning hubs, email and other services, 365 days a year, as essential in the increasingly competitive third level education sector, these outages are particularly worrying.

REQUIREMENT:

You are asked to address the following, based on a critical analysis of the above case study.

(a) Outline the main benefits and challenges for the college and, in particular, the Finance and Accounting department in implementing an ERP system. (8 Marks)

(b) Explain the advantages of using cloud computing services to implement an ERP system for the college. (8 Marks)

(c) Describe the core activities (stages) involved in a project to select, implement and deploy an ERP system in the college. Identify the main issues that need to be addressed during each activity/stage. (9 Marks)

[Total: 25 Marks]

Answers:

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[Total: 20 Marks]

[Total: 20 Marks]

Question 28

(a) Explain information asymmetry and how the Internet has reduced this. Give an example of where and how it has been reduced from a consumer's point of view. (6 marks)

(b) Describe four e-commerce revenue models and how they work, with examples of each. (6 marks)

(c) Outline an appropriate approach to take when planning a new e-commerce website for a business, and explain the main decisions involved in building the website. (8 marks)

[Total: 20 Marks]

Question 29

(a) Explain what is involved in a total cost of ownership (TCO) analysis of an organisation's technology assets. (7 marks)

(b) Define cloud computing, giving its main characteristics. Explain the different types of services it consists of, and the main benefits it offers to companies. (7 marks)

(c) Outline the concerns companies might have in relation to the use of cloud computing services. (6 marks)

Question 29

Hibernia Ltd., a steel manufacturing company, purchased an information systems package over a decade ago. The system focuses on daily transaction processing of stock, sales and purchasing and provides regular user summaries. Unfortunately, the output does not provide much of the necessary management information and as a consequence a Co-ordinator downloads output onto spread sheets which are then manipulated, enhanced and emailed to users. Managers and Senior Managers often complain that even this information is inadequate for their needs. In response, Hibernia has employed a business information consultancy firm to conduct a review of its information requirements. The consultancy firm reported that the existing system is outdated and much less functionally rich than more recent software on the market. The main issues and recommendations are summarised as follows:-

Purchase a new system

Hibernia should purchase a system (called InfoTek) which provides more complete, relevant and timely information. The InfoTek system is easy to understand and navigate and would offer a more

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comprehensive analysis than the existing package provides, including historical statistics and comparison to prices charged by competitors. InfoTek also offers potential for further customisation to local needs through its display functions and flexible formats. Users (Senior Managers, Managers and Others) would be able to use the software efficiently straight away without additional training. Daily transaction processing routines would remain unchanged.

Implementation of the new system

The system should be implemented by direct changeover because Hibernia's network servers have insufficient capacity to support both new and old systems. The installation should take place over a weekend to minimise disruption. Once installed, systems maintenance should be minimal.

Redesign of Co-ordinator's role

This role should be redesigned so that the existing post-holder becomes responsible for validating InfoTek output, co-ordinating information enhancements and providing support and advice to users. Hibernia's Managing Director is familiar with InfoTek through past working experience and agrees that it is a worthwhile package. She is, however, unhappy about how the consultants have conducted their review. In particular, she is concerned that very few Managers, Senior Managers and other information users have been involved in discussions. She also believes that any job design needs to concentrate on the characteristics of the post in order to gain maximum motivation and job performance. She therefore treats the findings and recommendations of the consultants with caution. A fellow Director has advised her not to implement the consultants' recommendations without further reference to the workforce as this might lead to staff resistance.

REQUIREMENTS:

(a) Discuss the reasons why there may be user resistance to the introduction of the new system and explain the types of support that could be organised to help users utilise InfoTek efficiently. (7 marks)

(b) Explain the issues Hibernia should consider before successfully implementing the recommended system. (9 marks)

(c) List and describe the options Hibernia has if they choose not to develop their software in house. (9 marks)

[Total: 25 Marks]

Question 31

(a) Discuss the reasons why it is necessary for businesses to invest so heavily in information systems and why there is such a high variability in the returns firms receive from their investment? (6 Marks)

(b) How can firms assess the value of information systems projects? (7 Marks)

(c) Describe the information system problems that result from poor project management. (7 Marks)

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[Total: 20 Marks]

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